

Houssam Yassin

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EDUCATION

PhD in Atmospheric and Oceanic Sciences (AOS) [Anticipated: Aug. 2022] 2017-present
 Princeton University, Princeton, New Jersey
 Advisor: Stephen Griffies
 Thesis: The geostrophic turbulence of boundary buoyancy anomalies

BSc with Honors: Mathematical Physics 2013-2017
 University of Alberta, Edmonton, Alberta

PUBLICATIONS

In review

Yassin, H., (in review.): The buoyancy staircase limit in surface quasigeostrophic turbulence. *Journal of Fluid Mechanics*. [arXiv:2206.03155](https://arxiv.org/abs/2206.03155)

Sutherland, B. R. & **Yassin, H.**, (in review): The nonlinear evolution of internal tides. part 2: Lagrangian transport by periodic and modulated waves. *Journal of Fluid Mechanics*.

Yassin, H. & Griffies, S. M., (in review): Surface quasigeostrophic turbulence in variable stratification. *Journal of Physical Oceanography*. [arXiv:2110.04242](https://arxiv.org/abs/2110.04242)

Peer-reviewed publications

Yassin, H. & Griffies, S. M., (2022): On the discrete normal modes of quasigeostrophic theory. *Journal of Physical Oceanography*, 52, 243-259. [doi:10.1175/JPO-D-21-0199.1](https://doi.org/10.1175/JPO-D-21-0199.1)

Yassin, H., (2021): Normal modes with boundary dynamics in geophysical fluids. *Journal of Mathematical Physics*, 64, 093102. [doi:10.1063/5.0048273](https://doi.org/10.1063/5.0048273)

Van den Bremer, T. S., **Yassin, H.** & Sutherland, B. R. (2019): Lagrangian transport by vertically confined internal wavepackets. *Journal of Fluid Mechanics*, 864, 348-380. [doi:10.1017/jfm.2019.30](https://doi.org/10.1017/jfm.2019.30)

AWARDS

Princeton Energy and Climate Scholar, Princeton University	2020-2022
Geophysical Fluid Dynamics Fellow, Woods Hole Oceanographic Institution	2019
Centennial Fellowship in the Natural Sciences, Princeton University	2017-2021
Dean's Silver Medal in Science, University of Alberta	2017
NSERC Undergraduate Student Research Award (<i>x3</i>)	2015-2017

OUTREACH & SERVICE

Co-presenter, Princeton Day School Energy and Climate Scholars	2021-2022
Student representative, AOS DEI Committee, Princeton University	2020-2021
Co-organizer, AOS student seminar, Princeton University	2019-2020
Co-organizer, AOS workshop on Machine Learning and Climate Modelling	2019

TALKS

Yassin, H. & Griffies, S. M., (April 2022): Surface quasigeostrophic turbulence in variable stratification. *Meeting of the Climate Process Team on Ocean Transport and Eddy Energy*. [In-person talk]

Yassin, H. & Griffies, S. M., (February 2022): Seasonality in surface quasigeostrophic turbulence with variable stratification. *Ocean Sciences Meeting 2022*. [Virtual talk]

Yassin, H. & Griffies, S. M., (Nov 2021): A submesoscale k^{-2} kinetic energy spectrum from surface quasigeostrophic theory with variable stratification. *74th Annual Meeting of the APS Division of Fluid Dynamics*. [In-person talk]

Yassin, H. & Griffies, S. M., (Sep 2021): Seasonality in surface quasigeostrophic turbulence with variable stratification. *NYU Courant Center for Atmosphere Ocean Science Student Seminar*. [Virtual talk]

Yassin, H. & Griffies, S. M., (Feb 2021): Discrete normal mode decompositions in quasigeostrophic theory. *Ocean Model Working Group Meeting for the Community Earth System Model*. [Virtual talk]

Yassin, H. & Griffies, S. M., (Jan 2021): Are the baroclinic modes incomplete? On the normal modes of quasigeostrophic theory. *2021 DRAKKER Annual Workshop*. [Virtual talk]

Yassin, H. (Nov 2020): Normal modes with boundary dynamics. *73rd Annual Meeting of the APS Division of Fluid Dynamics*. [Virtual recorded talk]

POSTERS

Yassin, H. & Griffies, S. M., (July 2022): Surface quasigeostrophic turbulence in variable stratification. *Boulder Summer School for Condensed Matter and Material Physics*.

Yassin, H. & Griffies, S. M., (June 2022): Surface quasigeostrophic turbulence in variable stratification. *Gordon Research Conference on Ocean Mixing*.

PROFESSIONAL MEMBERSHIPS

American Physical Society	Member since 2020
American Meteorological Society	Member since 2020
American Geophysical Union	Member since 2021